

King Khalid University
College of Computer Science
Department of Computer Science



جامعة الملك خالد
كلية علوم الحاسب الآلي
قسم علوم الحاسب الآلي

STUDENT LAB ACTIVITY REPORT

Course Name: Introduction to Machine Learning

Lab Teacher: Dr. AYMEN TRIGUI

Course Code: 462 CCS-3

Student ID: 445801958

Section Number: 904/907

S. No

Student Name: Abdulmalik Hussein Al-Haif

LAB ACTIVITY - 10

(1.5 Marks)

This lab aims to implement an SVM classifier of the data in the IRIS dataset.

Instructions:

- Load the IRIS dataset and understand its structure using statistical summaries and data visualization.
- Build two SVM classifiers of the dataset using two kernels: Linear Kernel and RBF Kernel.
- For the RBF kernel, vary C and Gamma to choose the most appropriate values for the Dataset Classification.
- Evaluate the performance of the classifiers.

Linear SVM accuracy: 1.0000

RBF SVM best configuration: C=10.0, gamma=0.1, accuracy=1.0000

Other tested RBF configurations:

C=0.1, gamma=0.01, accuracy=0.9000

C=0.1, gamma=0.1, accuracy=0.9000

C=0.1, gamma=1.0, accuracy=0.9667

C=1.0, gamma=0.01, accuracy=0.9000

C=1.0, gamma=0.1, accuracy=0.9333

C=1.0, gamma=1.0, accuracy=0.9667

C=10.0, gamma=0.01, accuracy=0.9667

C=10.0, gamma=0.1, accuracy=1.0000

C=10.0, gamma=1.0, accuracy=0.9667

Marks Obtained	/1.5
Teacher's Signature:	